

Esercizio.

Suppose B b.a. and denote τ the topology of $St(B)$. Consider

- $I : \tau \rightarrow \{\text{ideals of } B\}, Z \mapsto \{b \in B : N_b \subseteq Z\}$ and $X : \{\text{ideals of } B\} \rightarrow \tau, I \mapsto \bigcup \{N_b : b \in I\}$. Then I, X are one the inverse of the others, so there is a correspondence between open subsets of $St(B)$ and ideals of B . Furthermore $U \subseteq St(B)$ is open if and only if $X(I(U)) = U$.
- $F : \tau \rightarrow \{\text{filters of } B\}, Z \mapsto \{b \in B : N_b \supseteq Z\}$ and $X : \{\text{filters of } B\} \rightarrow \{\text{closed subsets of } St(B)\}, F \mapsto \bigcap \{N_b : b \in F\}$. Then F, X are one the inverse of the other, so there is a correspondence between closed subsets of $St(B)$ and filters of B . Furthermore $C \subseteq St(B)$ is closed if and only if $X(F(C)) = C$.

Dimostrazione.

- Let $Z \subseteq St(B)$. Then $X(I(Z)) = X(\{b \in B : N_b \subseteq Z\}) = \bigcup \{N_b : N_b \subseteq Z\} = Z$, where the last inequality holds because Z is open. Let J ideal of B . Then $I(X(J)) = I(\bigcup \{N_b : b \in J\}) = \{c \in B : N_c \subseteq \bigcup \{N_b : b \in J\}\}$. Then it is clear that $I(X(J)) \supseteq J$. For the converse, let $c \in I(X(J))$. Then $N_c \subseteq \bigcup \{N_b : b \in J\}$. Since we know $St(B)$ is compact and N_c is closed, we have N_c compact, so we can extract a finite covering: $N_c \subseteq \bigcup_{i=1, \dots, k} N_{b_i} = N_{\bigvee_{i=1, \dots, k} b_i}$. So we deduce $c \leq \bigvee_{i=1, \dots, k} b_i \in J$ and since J ideal we have $c \in J$ as well. This way we have incidentally proved that if $U \subseteq St(B)$ is open then $X(I(U)) = U$. For the converse observe that $X(I(U))$ is union of open subsets of $St(B)$.
- Pretty much the same.

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